M.E. MODEL PROJECT REVIEW • PROVISIONAL STRUCTURE

# AVA Evaluation

An Empirical Study of Conflict Resolution for Assertional State in Multi-Agent AI Systems



OWL ANALOGY

**Agree · Verify · Abstain**

Proposal/review aid — not an official KEC template, thesis chapter, or implementation report.

# Eight provisional modules — a route through the proposal

**01 Problem and background**

Why shared meaning can fail even when replicas agree

**02 Scope and state model**

Separate additive facts from single-valued assertions

**03 Relation classes**

Classify the relationship before choosing a policy action

**04 Agree / Verify / Abstain model**

Keep policy actions distinct from terminal outcomes

**05 Research questions and comparison arms**

Compare strong baselines fairly, and separate classifier from policy

**06 Dataset, annotation and metrics**

Build a trustworthy gold set; report safety errors separately

**07 SCP-MA experiment and convergence target**

Use a minimal fault simulator only after the research gate

**08 Risks, limitations and staged plan**

Protect the research gate and keep the M.E. scope manageable

# Replicas can agree while meaning stays unsafe

**Agent A**

database = PostgreSQL

**Agent B**

database = DynamoDB

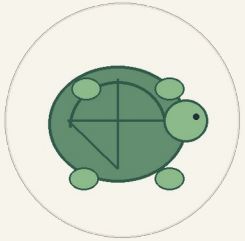
**Both events reach every replica**

**What AVA asks**

Are the claims paraphrases, compatible, soft conflict or hard conflict?

Should the system agree, verify or abstain?

## Mechanical replication ≠ semantic correctness



TURTLE ANALOGY: pause before merging

# Scope: additive state versus assertional state

## A • Additive state

Logs, observations, discovered entities
Mergeable facts; standard G-Set / OR-Set CRDTs
Not claimed as novel

## Q • Assertional state

Single-valued fact per declared key
Competing values may conflict
Primary research object



BEE ANALOGY: separate roles

**Assertion q = (key, value, source agent, causal metadata, logical timestamp)**

Same-key scope must be explicit: key type, role, temporal scope, and cardinality.

Out of scope: Byzantine agents · permanent partitions · cross-key intent conflicts · general open-domain NLI

# Four relation labels — before policy actions

## Paraphrase

Same fact, different wording

PostgreSQL is selected / Project uses PostgreSQL

## Compatible

Both can be true in declared scopes

Transactions use PostgreSQL / Sessions use DynamoDB

## Soft conflict

Different claims; policy/time may resolve

API v1 active / migrating to API v2

## Hard conflict

Mutually exclusive for same key/time

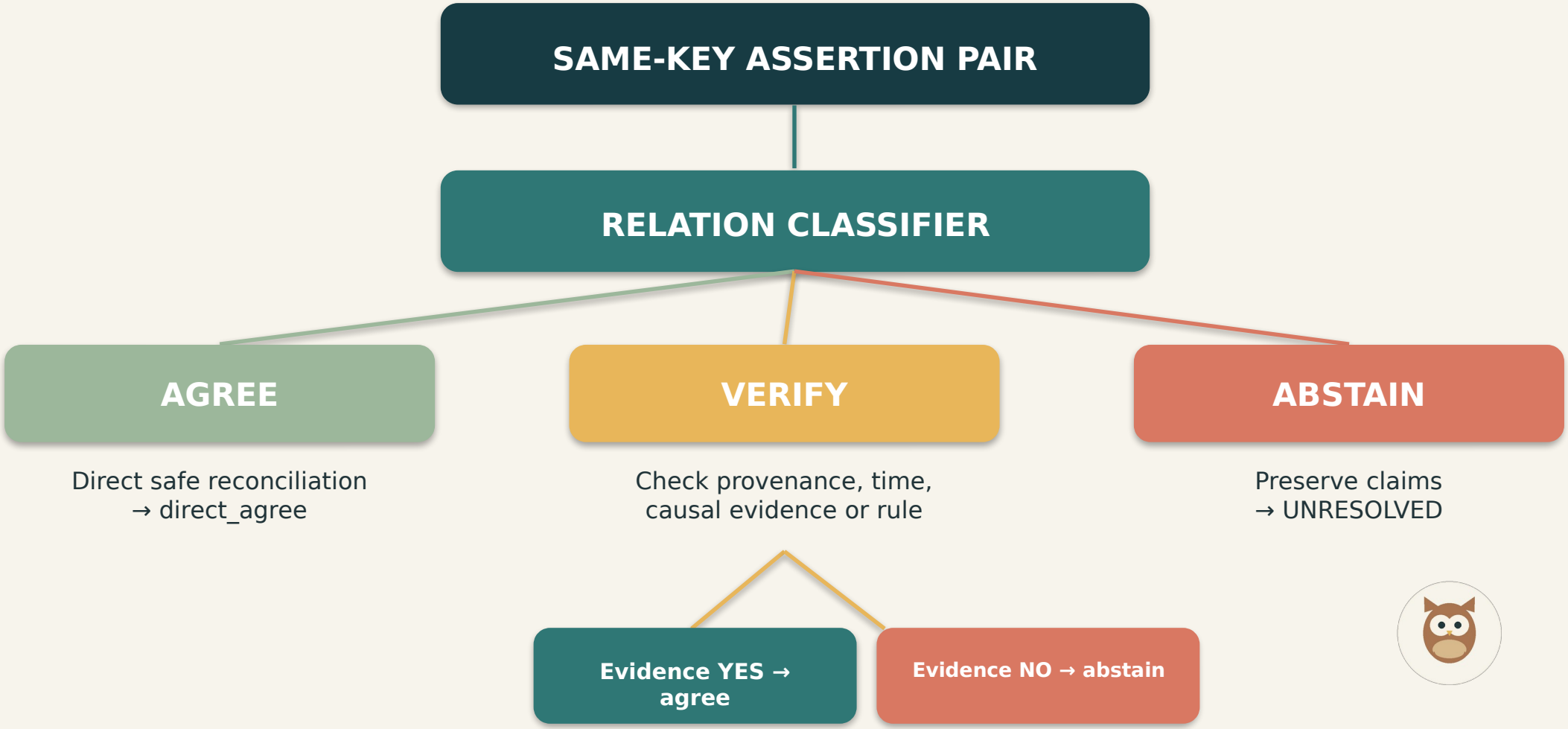
database=PostgreSQL / database=DynamoDB

**Label the relationship first. Choose system action second.**



**FOX ANALOGY: notice subtle differences**

# AVA policy: Agree, Verify, or preserve as Unresolved






OWL ANALOGY

# Seven arms — two different layers to compare

## RELATION / CLASSIFICATION SIGNALS

| | |
|---|---|
| **B2 Embedding** | similarity |
| **B3 NLI** | entail / contradict / neutral |
| **B4 Hybrid SCS** | weighted semantic signals |
| **B5 LLM Judge** | fixed prompt/model; repeated trials |
| **B6 Jev** | structured System-One questions |

## RESOLUTION / SAFETY POLICIES

| | |
|---|---|
| **B1 Deterministic** | argmax(timestamp, agent_id) |
| **B7 Preserve / abstain** | non-identical remains unresolved |
| **AVA Verify stage** | evidence-based intermediate check |

**Fairness rule: report classifier accuracy separately from final safety and resolution outcomes.**

# Dataset: human-labelled pairs, context and evidence

1. **Controlled synthetic**
   balanced edge cases
2. **LLM-generated realistic**
   human checked
3. **Real traces (if feasible)**
   legal/practical access

## Record fields

Declared key + type · assertion A/B · context/time
Evidence/provenance · gold label · difficulty

## Gold-label process

3 annotators → agreement statistic → adjudication
LLM output alone is not ground truth



BEE ANALOGY

# Safety metrics stay separate — no misleading composite

### FMR • unsafe agreement

Wrong Agree on actual conflicts (direct or verified) ÷ actual conflicts

### FCR • missed safe resolution

Failure to resolve actual Paraphrase + Compatible cases ÷ those resolvable cases

### Unnecessary abstention

Subset of FCR after Verify on resolvable cases; reported separately too

**Report a safety–coverage picture, not one “winner” score.**

Also: per-class F1 · calibration (ECE / Brier) · latency · token / compute cost · throughput



TURTLE ANALOGY

# SCP-MA: test decisions under asynchronous event delivery

### Agents

concurrent writes

### Immutable log

event IDs + causal metadata

### Network

delay · duplicate · reorder · disconnect

### Projection

resolve value or UNRESOLVED

**duplicate delivery**

**reordered delivery**

**delayed delivery**

## Measure

Convergence rate/latency · divergence · bytes/messages · duplicate handling · recovery

## First validation

Semantics-free deterministic simulator before semantic methods are connected



PENGUIN ANALOGY

# Convergence is a target condition — not yet proved

**Same valid event set E + deterministic Resolve(Canonicalize(E))**
**⇒ identical projected state at every replica**

### Canonicalize

same event set

### Resolve

pure deterministic function

### Project

same value or UNRESOLVED

**Open proof obligations: canonical inputs · tie rules · deterministic Verify · exact meaning of “state”**

# Prior work narrows broad claims

| | | |
|---|---|---|
| **Pair relations / false merges** | **MELD** | 104 pairs; distinct keys |
| **Conflict-preserving state** | **StateFuse · TOKI** | history, provenance, projection, abstention |
| **Long-history relational QA** | **SubtleMemory · TANGLE · STALE** | downstream memory tasks |
| **Coordination / governance** | **S-Bus · Governed Shared Memory** | structural concurrency / supersession |
| **NLI / evidence pairs** | **ECon · RefNLI · ANLI** | adjacent pair reasoning; different setup |

**Interpretation: substantial overlap exists; novelty remains unresolved pending full source-wide verification.**



# State the literature numbers with their caveats

### MELD

104-pair set uses distinct claim keys

### CodeCRDT

5–10% from manual 60/600 sample

### SCF

27.9% precision · 65.2% recall · F1 39.0

### Freshness pipeline

+10.8 / +21 pp are whole-pipeline; executor-only +2 / 0

**Partial findings challenge scope; they are not a final novelty verdict.**

# Key decisions before any build

## Key schema

Cardinality, type, role, temporal scope

## Label guide

Soft vs hard; pair vs context

## Layer boundary

Classifier output → policy action mapping

## Evidence for Verify

Allowed deterministic sources and rules

## Feasibility

Annotators, sample size, compute/API budget

## Requirements

Supervisor, college format and milestones unknown



OWL ANALOGY: ask before assuming

# Staged plan — preserve the research gate

**A**

## Verify literature

full texts, appendices, active disconfirmation

**B**

## Settle design

key schema, labels, policy mapping, feasibility

**C**

## Core benchmark

only if gate survives; human gold labels

**D**

## Minimal simulator

validate deterministic event projection first

**E**

## Decide scope

retain / narrow / reframe / abandon

**Current gate: Pass 3 incomplete · novelty unresolved · implementation and formal drafting not authorized**

THE STUDY IN ONE SENTENCE

# When agents disagree about the same state key, when is it safe to agree, when should the system verify, and when should it preserve uncertainty?



FOX ANALOGY — question carefully

**Measure it. Don’t assume the intelligent method is better.**

Research proposal only · no result claimed · novelty and proof obligations remain open

# A hypothetical AI-agent team shares one key

The labels are illustrative only: no real identity, capabilities, integrations, roles or performance are implied.

**KEY  service.demo-api.primary_db**

Scope: demo-api • current release window • database enum • exactly one value

| Grok bot | GPT dots | Claude duet | Harkpro | Wajo Fo | Wabi |
|---|---|---|---|---|---|
| hypothetical write | hypothetical write | hypothetical write | hypothetical write | hypothetical write | hypothetical write |
| **PostgreSQL** | **PostgreSQL** | **PostgreSQL** | **DynamoDB** | **DynamoDB** | **DynamoDB** |

**3 candidate writes**

primary_db = PostgreSQL

**3 candidate writes**

primary_db = DynamoDB

**Same declared key + scope + interval: equal values may be equivalent; the two values cannot both occupy a single-valued key.**

# Do not collapse four different questions

In the example, disagreement is not “solved” by counting agents or choosing the last arrival.

## 1 PROVENANCE

**Six hypothetical labels**
**Each write stays attached to its source.**

Who said which value?

## 2 RELATION

**Same-key / same-scope**
**PostgreSQL vs DynamoDB = hard-conflict candidate.**

Can both be true?

## 3 POLICY ACTION

**VERIFY**
**Only with a named evidence source + deterministic rule.**

What should happen?

## 4 RESULTING STATE

**Evidence not supplied**
**→ UNRESOLVED; keep claims + provenance.**

What gets published?

**TURTLE ANALOGY: pause when evidence is missing — teaching analogy only, not a protocol component.**

# Key design points to get right before any build

These are open design questions, not claims that the design is already settled.

### Same-key contract

Key, type, role, time window, cardinality

### Layer separation

Provenance → relation label → action → output state

### Verify rule

Allowed evidence, freshness, deterministic tie behavior

### Fair comparison

Same inputs; classifier scores separate from policy outcomes

### Safety first

False merges, false conflicts, abstention, calibration, cost

### M.E. feasibility

Prioritize a small core; stage simulator/scaling

## Likely questions

Why not newest-wins? Arrival order is not truth. • Does Verify guarantee correctness? No. • Is novelty established? No—Pass 3 is incomplete.

# Monday 26 October — ready-to-present checklist

No review time or official KEC module titles were supplied; confirm those with the reviewer.

## Before the review

- [ ] Open the PDF once; check slide order and diagrams on the presentation device.
- [ ] Keep PPTX + slide PDF together; keep report PDF as backup/reference.
- [ ] Rehearse the 90-second opener and the one-key scenario.
- [ ] Call the six names hypothetical labels only; assign no real capabilities.
- [ ] State the gate honestly: Pass 3 incomplete, novelty unresolved, no implementation.
- [ ] Ask which official modules/template and next milestone the college expects.

## Open with this

“Same events reaching all replicas does not guarantee safe meaning. I’ll test how same-key disagreement should be classified and when to resolve, verify or preserve uncertainty.”

## Ask for feedback

Is the key/scope precise?
Which work should be core vs staged?
What must pass before implementation?

# Pass 3 evidence • close overlaps

- **Same-key commit and memory policy:** MemTX v2: evidence-bearing same-slot conflict gate; 90 cases / 5 backbones. Shared store, not async replica convergence.
- **Deferral is prior art:** TARL v5: append / noop / revise / reject_conflict / defer_verify; 5,422 cases with gold next state.
- **Classify then reconcile:** LatticeMind v1: Credibility vs Coordination; 73/75 on a 75-case ConflictBank set, one model.
- **Sources:** arXiv:2607.23929v2 · 2608.03699v5 · 2608.08236v1

# Async delivery • measure what converges

- **Direct async example:** Governed Shared Memory v1: supersession 90/90 conditional on both writes landing; 98/200 overall.
- **Admission matters:** A synchronous dedup gate rejected 206/400 writes before the asynchronous detector; report gate and detector separately.
- **Different system questions:** CoAgent repairs plan invalidity; S-Bus rejects structural stale-read/write races; neither is same-key truth adjudication.
- **MELD caution:** Status convergence can coexist with order-sensitive live sets / retrieval: 112/200 orderings changed the live set.

# Strong overlap ≠ exact experiment

- **StateFuse / TOKI:** Conflict-preserving replicated history, projection/abstention, provenance, time-aware policies and confirmation already appear.
- **StateFuse caveat:** Table 3 (282): conservative 64.9% accuracy / recall 100% / false-certainty 2.1%; aggressive 97.5% / 100% / 2.5%. Table 4 (50): success 80%→100%, false-certainty 20%→0%, 2.4 calls, 0 cost; controlled, not async convergence.
- **SCF scope:** 600 simulated workflow runs; 27.9% precision / 65.2% recall / 39.0% F1 / 100% completion. Intent prevention ≠ claim convergence.
- **Survey / code benchmark:** Always-On: scoped 435-work survey, not exhaustive. CodeCRDT 5–10% = preliminary 60/600 manual sample.

# Verification needs context

- **Evidence-pair errors:** ECon: weak-model recall limits; resolution may be unjustified or prior-informed.
- **Reference mismatch:** REF-NLI: 1,143 claim/evidence pairs, including ambiguous reference/context cases.
- **Adversarial NLI:** ANLI v2: ~19k / 47k / 103k examples; dev/test checked, training not fully verified.
- **Abstention is not verification:** Varshney et al. (17 datasets): no method consistently beats MaxProb across IID/OOD/ADV.

# Gate stays open • no novelty verdict

- **Broad novelty is unsafe:** Classification, evidence-aware handling, defer/abstain, conflict preservation and async supersession each have close prior art.
- **Candidate comparison remains untested:** Same-key relation + evidence-based Verify vs fixed-winner vs preserve-unresolved under async reordering, with defined convergence outputs.
- **WH-1/2/3:** Exact authoritative definitions are missing; leave undefined and obtain the source wording.
- **Scope of this appendix:** Targeted excerpts, not full source/appendix audits or independent reproduction. Pack remains provisional; no implementation or convergence proof claimed.